

PRODUCT WHITEPAPER

DecisioWorks

Production Decision Capability Toolkit

Verifiable capability chain for planning and scheduling, intelligent production scheduling, data asset formation and MES/ERP integration

core positioning

DecisioWorks is a capability toolkit for manufacturing production decision scenarios, providing enterprises and ecosystem partners with a complete capability chain from data access, planning and scheduling to solution comparison, organizational review and continuous optimization.

In other words: DecisioWorks helps companies transform production data into runnable, verifiable, and reviewable plans and decision capabilities. Enterprises can start with a specific scenario and then gradually expand according to actual needs after running through it.

Core capabilities are not just about “making good plans”, but also allowing plans to enter into goal and rule governance, plan comparison, organizational review, plan linkage and feedback iteration to further support enterprises in “making good decisions”.

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Commercial Edition: <https://gitee.com/qiaoyx-or/decisioworks>

Research Edition: <https://gitee.com/qiaoyx-or/decisioworks-research>

Product Home: <https://qiaoyx-or.github.io/production-decision-sandbox>

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1. Product Overview

DecisioWorks is a basic capability toolkit for manufacturing production decision scenarios. It mainly serves scenarios such as production planning, intelligent production scheduling, capacity analysis, data asset formation, operational validation and system integration. It is oriented to existing ERP, MES, WMS, industrial Internet platforms and data governance systems to enhance production decision capabilities, rather than replacing existing business systems of enterprises.

The core value of DecisioWorks is to allow manufacturing companies and ecosystem partners to start from a qualified data scenario and complete the minimum closed loop from data access, business semantic expression, goal and rule configuration, GOCK controlled solving, to result analysis, organizational review and plan linkage. Enterprises can verify first and then decide whether to expand; partners can embed first and then form an industrialized solution.

Product positioning

DecisioWorks separates production decision capabilities from one-time project delivery and forms a verifiable, integrable, licensable, and sustainably evolving toolkit capability.

2. Background and pain points

Many manufacturing companies already have ERP, MES, WMS, equipment systems and a large amount of business data. However, production planning, job scheduling, order response, material shortage processing, equipment exceptions and delivery trade-offs are still highly dependent on manual experience. The real difficulty is not “whether there is a system”, but whether the system can organize data, business, models, solutions and feedback into a stable decision chain.

If one of the following applies to your business, DecisioWorks can serve as a lighter, more verifiable starting point for a traditional large APS implementation.

Common pain points	Specific performance	DecisioWorks entry point
Data semantics are not unified	The fields exist, but the relationships between orders, processes, resources, calendars, and inventories are unclear.	Computable and reviewable data assets are formed through standardized data interfaces.
System implementation risks are high	Traditional projects are often invested first and verified later, and the results are not revealed until they go online.	Run verification from small scenarios first and move the risk forward to the data access stage.
Manual adjustment destroys the closed loop	When the scheduling results are not trustworthy, return to Excel and modify them manually, and the constraint relationships will be broken up.	Use goal and rule governance, result analysis and feedback mechanisms to support organizational review.
Capabilities are difficult to reuse	Project rules and ad hoc scripts are prone to failure after scene changes.	Retain the stable parts as model capabilities, interface specifications and scenario templates.

3. Product positioning and ecosystem roles

DecisioWorks is the production-decision capability base. It provides stable, common, and reusable toolkit capabilities to help customers and partners connect the key links between data, models, solutions, and business applications, and collaborate with existing systems and processes of the enterprise.

Ceju focuses on toolkit research and development, capability licensing and product support. DecisioWorks provides standard actions, capability orchestration mechanisms, data templates and sample recipes; enterprises or ecosystem partners can configure industry processes, customer rules and special templates accordingly to organize system integration, on-site implementation and long-term operation and maintenance.

Capabilities provided by DecisioWorks	Application methods for enterprises and partners
Provide standardized data interface, goal and rule governance, controlled solving, process orchestration, result analysis and plan linkage capabilities.	Embed ERP, MES, WMS and other business systems to enhance production decision capabilities.
Help enterprises transform real data into computable, reviewable, and executable decision inputs.	Start verification with specific data scenarios and then gradually expand to more business links.
As a component of production decision capabilities in MES/ERP, consulting solutions and system integration projects.	Combined with partners' customer base, industry knowledge and on-site services, a complete application solution is formed.
It supports understanding the boundaries of capabilities from the research version, and then entering the commercial version for authorization and formal verification.	Select the corresponding version and licensing plan based on learning assessment, business application or partner integration needs.

4. Core architecture and capability chain

The core of DecisioWorks is not a certain scheduling algorithm, but a set of layered architecture between existing business systems and organizational decision. It converts on-site data into a computable structure, and GOCK handles model construction and optimization, and then enters organizational decision through result analysis, plan review, plan linkage and feedback iteration.

4.1 Layered architecture: the production-decision capability layer on top of the existing system

DecisioWorks uses "Scenario × Business × Model × Solver" as the top-level logic, and is collaboratively implemented by DataSets, DecisioCore, GOCK and Web Cockpit. DataSets provides scenario data, DecisioCore organizes the decision chain, GOCK handles model construction and optimization, and Web Cockpit provides operation and display. Each component collaborates through data, model calls and result feedback to form a complete link from business input to organizational decision.

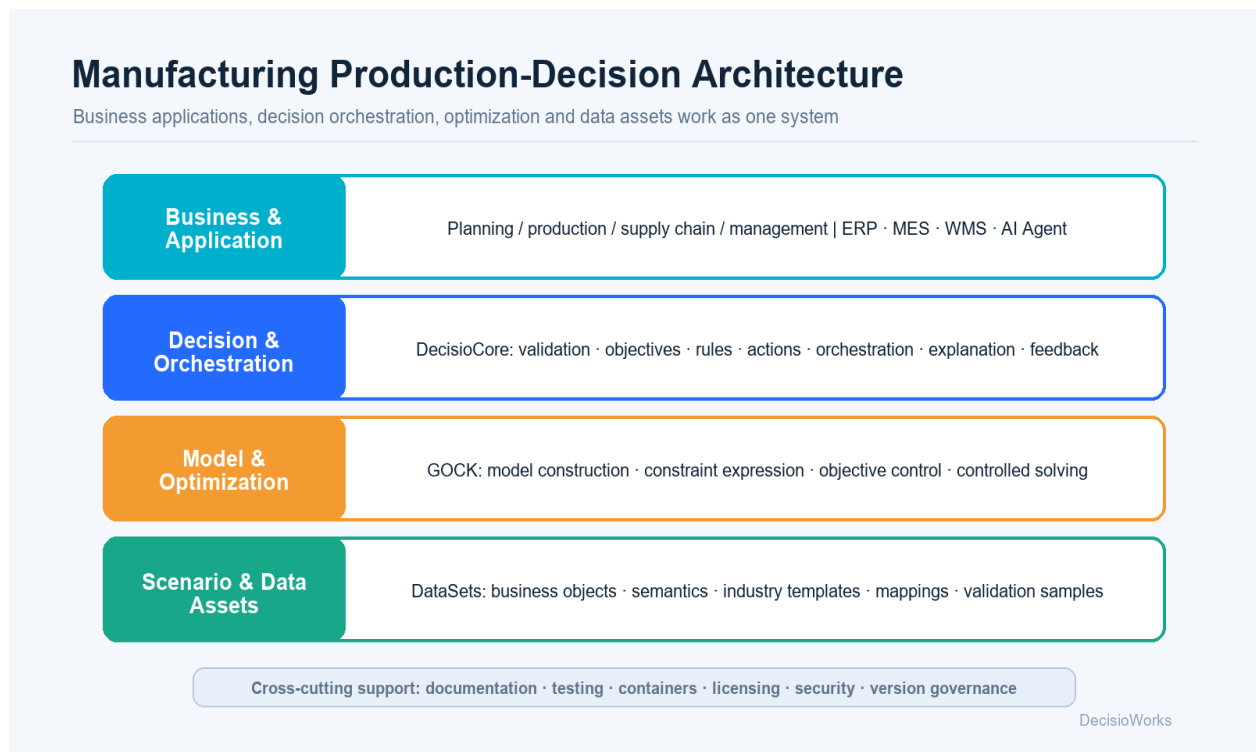


Figure 1 The relationship between DecisioWorks software architecture and controlled calculation

Architecture level	core competencies	formative effect
Organizational decision and execution level	Plan comparison, manual review, choice confirmation, plan release and execution commitment	Organize and complete plan selection and implementation arrangements based on on-site conditions
DecisioWorks Decision application layer	Scenario configuration, goal and rule governance, process orchestration, result analysis, plan linkage and feedback	Transform solver results into interpretable, reviewable, and executable decision objects
DecisioCore Decision organization and integration layer	Data verification, goals and rules, standard actions, model adaptation, result analysis and plan feedback; provide source code open reference implementation	Lowering the integration barrier for manufacturing enterprise IT teams and industrial software partners
GOCK calculation layer	Handle model construction, mathematical expression of goals and constraints, optimization calculation and controlled operation	Handle complex mathematical derivation and form stable and reusable optimization computing capabilities
Data semantics and field layer	The source system provides business facts, DataSets provides scenarios and standard data instances, and DecisioCore verifies and maps business objects.	Form computable data assets and receive execution deviations and field feedback

core structure

Component collaboration: DataSets → DecisioCore (validation, target rules, orchestration) → GOCK (model and optimization) → DecisioCore (analysis, interpretation, feedback) → Web Cockpit / API / Next round of planning.

Web Cockpit is the operation and display entrance, and business personnel can review the plan and promote implementation accordingly.

It is not a one-way pipeline of "data input, schedule output", but a closed loop of "field data - model calculation - plan explanation - organizational decision - execution feedback".

4.2 Capability closed loop: from data entry to execution feedback

The hierarchical architecture solves "where to place capabilities", and the capability chain solves "how capabilities can be continuously run". After the two are combined, production decision can be transformed from a one-time solution into a process that can be inspected, reviewed, linked, and continuously evolved.

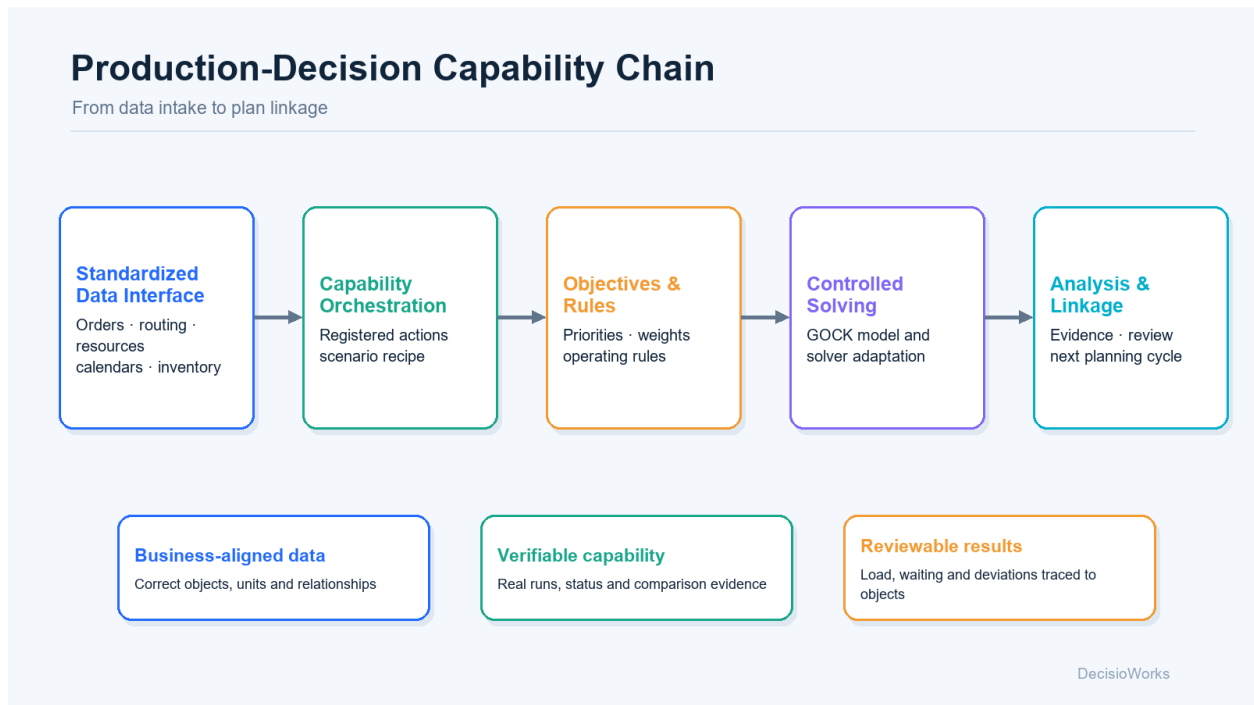


Figure 2 DecisioWorks production-decision capability chain

Capability chain	illustrate	form value
data entry	Objects such as orders, products, materials, BOM, processes, resources, calendars, and inventory enter standardized interfaces.	Form computable input and reduce semantic breaks after field alignment.
business expression	Site objects, resource capabilities, process relationships, time calendars, and status relationships are explicitly expressed.	Let the on-site language be transformed into a structure that the model can handle.
Governance by goals and rules	Target preferences and business rules such as delivery time, capacity, inventory, model change, load, etc. are configured uniformly.	Enable business choices to be discussed, adjusted and reviewed.
model organization	Variables, relationships, constraints, goals and rules are unified by the model layer.	Retain stable capabilities and avoid re-stacking temporary logic for each project.
GOCK controlled solving	Organize calculus within authorization, time, resources, input consistency, and operational rules.	Develop stable and manageable model-based solving capabilities.
Result analysis	Present load, delivery time, bottlenecks, conflicts, constraint hits, plan differences and adjustment basis.	Help organizations understand results rather than passively accept system answers.
Organization review and plan linkage	Hand over the candidate plans to planning, production, supply chain and other roles to complete the selection and enter into plan release or system linkage.	Let the calculation results enter the real decision and execution process.
feedback iteration	Bring execution deviations, manual adjustments, rule changes, and next round of data back into the capability chain.	Allow production decision capabilities to continuously calibrate and evolve.

5. Standardized data interface and data asset formation

The first level of value of DecisioWorks is to help enterprises form high-quality production decision data assets. It's not that many companies don't have data, but that the data doesn't enter unified business semantics; it's not that there are no fields, but that there is a lack of calculation relationships between fields for planning, scheduling, capacity, and delivery.

The value of a standardized data interface is not to align fields, but to organize orders, products, materials, BOMs, processes, resources, calendars, inventories, constraints, and goals into computable, traceable, and reusable business assets.

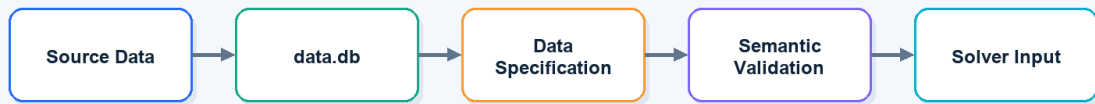
data object	effect
Order/Requirement	Express customer needs, internal planning needs, lead times, priorities and quantities.
Product/BOM/Material	Express product structure, material dependencies, process-level material requirements and set relationships.
Craft/Resources/Calendar	Express machining paths, work center capabilities, shifts, maintenance schedules, and available times.
Inventory/Constraints/Targets	Express inventory boundaries, business rules, hard constraints, soft constraints, and goal preferences.
result field	Agree on the fields required for plan result return, analysis interpretation and subsequent review.

The progressive value of data asset formation

The first step is to form computable data assets; the second step is to test the suitability of the data and the model through actual operation; the third step is to have business personnel review the plan before entering the decision process. Data can be solved and results can be accepted separately, and asset formation, operational validation, and result application are progressed in the same link.

From data.db to a Computable Business Model

Data structure is only the start; business semantics and relationships must also close



A data specification defines how business facts enter the decision chain; it does not replace the business itself.

DecisioWorks

Figure 3 From Raw Data to a Computable Data Specification

Data acceptance focuses on ContractReady (structure), RelationReady (relationship), ModelReady (model), SolverReady (solving conditions) and DecisionReady (result interpretation and business review) in order, gradually advancing from data quality to solution availability.

6. GOCK and Model-Data Resonance

6.1 The position of GOCK in the core architecture

GOCK (general-purpose optimization calculus kernel) is the core calculation layer of DecisioWorks. DecisioWorks provides users with data access, scenario configuration, process orchestration, result analysis and plan linkage; DecisioCore provides an open reference implementation and integration entry point; GOCK handles model construction, algorithm adaptation and solver execution within a controlled boundary.

The key significance of GOCK is not to encapsulate a certain scheduling algorithm, but to separate business expression, model organization and algorithm execution. On-site changes mainly enter through data, goals and rules; stable relationships are retained as model capabilities; different algorithms and solvers are adapted and invoked under a unified boundary. This avoids solidifying project experience into scripts that are difficult to maintain, and avoids stuffing all complex judgments into unexplainable black boxes.

GOCK capabilities	assume content	architectural significance
model organization	Organize business objects, relationships, constraints, goals, and rules into computable structures.	Enable models to be reused across scenarios rather than relying on one-off project scripts.
Algorithm adaptation	Decouple the model from specific algorithms and solvers, and unify input, output, and running context.	Leave room for choice for different problem sizes and solution strategies.
controlled solving	Invoke computing capabilities within authorization, time, resources, parameters, and input consistency.	Implement authorization and operating conditions, return identifiable status, and support operation management and exception handling.
Result delivery	Returns planning results, status information, and context that can be used for analysis and review to higher levels.	Allowing results to be interpreted, compared, and fed into the subsequent decision process.
Ability accumulation	Retain stable mathematical relationships into core capabilities, leaving changes to data and configuration.	Support toolkits continue to expand without starting over with each site.

6.2 Model-Data Resonance: continuous calibration of data and models

In terms of responsibilities, DecisioCore's objective_system and marginal_control organize target preferences and on-site rules, engine_adapters is responsible for model call adaptation; GOCK takes over its mathematical expression and optimization calculation. Interpretation and feedback after solution are returned to DecisioCore.

The "module" in "Model-Data Resonance" refers to the model, and the "number" refers to the data. Data is not as simple as a table input and a result output; it needs to enter the model with semantics such as orders, processes, resources, calendars, inventory, status, and business rules. Models are also not static formulas, but computable structures that can take on-site relationships, target preferences, and rule conditions.

The data and model are not a one-in and one-out calling relationship, but a symbiotic relationship that is repeatedly calibrated in the same capability chain: data enters the model with business semantics, the model inherits the on-site relationship with a computable structure, and the results are returned to the organizational decision through analysis and feedback.

Visible simplicity, invisible complexity

What users can see is accessing data, configuring scenarios, running solutions and viewing results; what users cannot see are standardized interfaces, model organization, target systems, controlled solvings and feedback iterations. DecisioWorks Keep the complexity in the model and leave simplicity to the users.

7. From “making good plans” to “making good decisions”

Traditional scheduling systems often regard "generating a feasible plan" as the end point. DecisioWorks further views the plan as a key object in the decision process: it needs to be able to be interpreted, compared, reviewed, adjusted, released, and receive feedback after execution.

"Planning well" focuses on how to form a feasible and better-performing arrangement given data, resources, constraints and goals; "Making decisions well" further answers why the plan is formed in this way, what are the alternatives and their costs, and how to complete the choice and implementation based on the on-site situation.

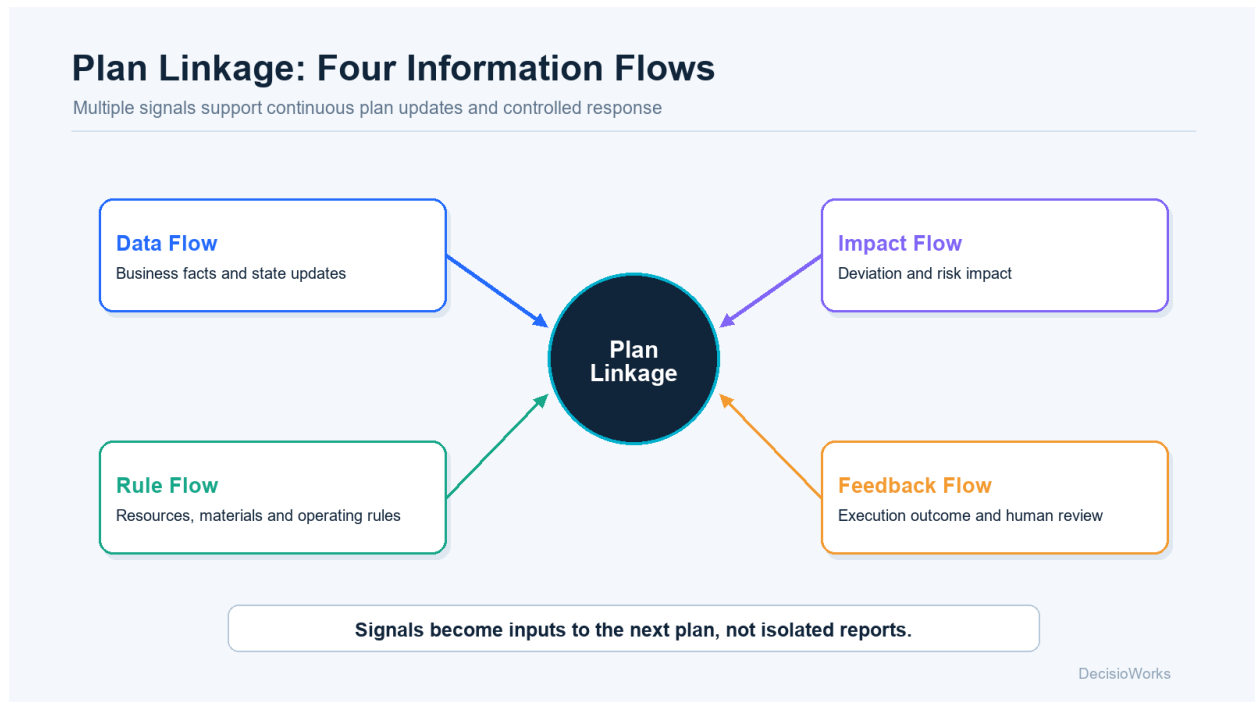


Figure 4 Plan Linkage: Four Information Flows

Dimensions	Make plans	Make good decisions
core issues	Whether a feasible plan can be formed under order, process, resource, calendar and inventory conditions.	Why the plan is formed in this way, whether it is worthy of acceptance, and how to choose and implement it.
main input	Production data, resource capabilities, process relationships and basic constraints.	Production data, as well as target priorities, business rules, risk appetite and site status.
main output	Scheduling results such as planning time, job sequence, resource allocation, etc.	Comparable options, bottlenecks and conflicts, impact description, adjustment basis and plan linkage results.
system role	Calculate and generate feasible or optimized plans.	Processing the calculable part reduces organizational judgment, comparison and review costs.
organizational role	Receive the results and make manual adjustments if necessary.	Complete plan selection, approval and implementation arrangements based on on-site conditions.
Closed loop method	One-time generation or periodic rearrangement.	Run-review-publish-execute-feedback-recalculate.

core judgment

DecisioWorks further transforms plans into decision objects that are comparable, explainable, reviewable, and linkable. The system provides the basis for calculation and analysis, organizes decisions based on on-site conditions, and promotes implementation.

From generating plans to adopting solutions, DecisioWorks clearly organizes calculation basis, plan differences and impact relationships, allowing planning, production and supply chain personnel to make collaborative decisions around the same set of information.

8. Applicable scenarios and typical objects

The applicable scope of DecisioWorks is not limited to the following scenarios, but these scenarios are most suitable as early validation entry points. Their common characteristics are: data can be accessed, questions can be explained clearly, and results can be reviewed by business personnel.

object/scene	Applicable method	value
Manufacturing companies that want to form high-quality data assets	Organize orders, processes, resources, calendars, inventory, and constraints into standardized interfaces.	Get data from dispersed records into computable production decision structures.
Companies that are planning APS or smart production scheduling but have unclear directions	Run verification first with real or masked data before judging ROI and construction paths.	Reduce project risks of investing first and verifying later.
MES/ERP/WMS Manufacturer	Embed DecisioWorks into the existing system solution as a production-decision capability component.	Complement planning and scheduling, goal expression, constraint solving and result interpretation capabilities.
Consulting agencies and system integrators	Add data diagnosis, effect evaluation, result verification and ROI judgment in the plan stage.	Turn consulting recommendations into actionable evidence.
Park and public service platform	Organize enterprises to carry out low-risk verification from qualified data scenarios.	Form a service hub for intelligent digital transformation, industrial software and AI+ manufacturing.

The common features of these scenarios are: data can be accessed, problems can be explained clearly, and results can be reviewed by business personnel; it is not required to cover the entire factory and process at the beginning.

9. Implementation path and pilot verification

Enterprises or partners can start with a light pilot, establish a baseline, adjust recalculations, and review the results to form operational evidence that supports business acceptance, subsequent integration, and continued operations.

The goal of the pilot is not to cover the entire factory at once, but to use a scenario with clear boundaries and accessible data to verify whether the capability chain can be closed.

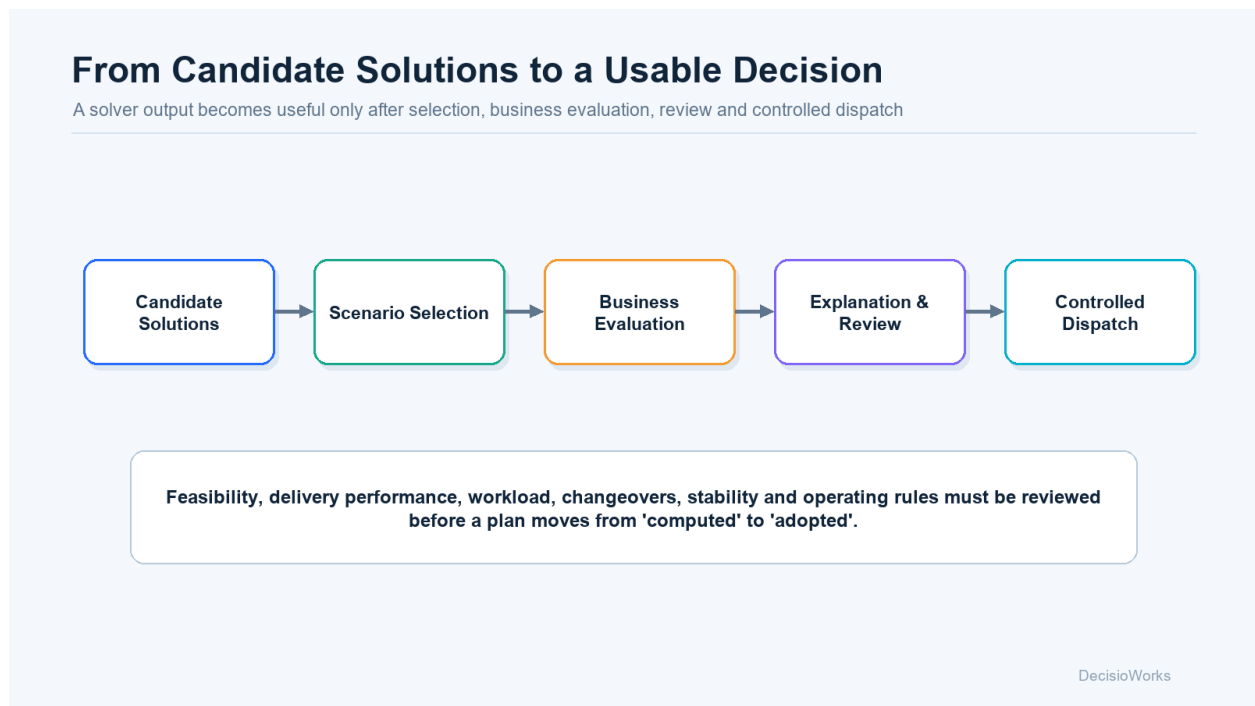


Figure 5 From candidates to available decisions

1. Scenario definition: clarify the pilot objects, business boundaries, inputs and outputs, and success criteria.
2. Data mapping: Organize sample data into standardized decision inputs and confirm field semantics.
3. Goal and rule configuration: Express business intentions such as delivery time, capacity, inventory, model change, load, etc., as well as necessary on-site rules into computable configurations.
4. Baseline and adjustment recalculation: Save the current plan and evaluation criteria, and run the solver within the same data range; adjust a small number of goals, rules, or resource conditions in each round to compare differences in plans.
5. Business review: Explain bottlenecks, conflicts, constraint hits, plan differences and costs, and the responsible person will confirm the continuation, adjustment or stop, and record the reasons for acceptance or modification.
6. Template retention and expansion: save the mapping, configuration, rules and processes that have passed the verification, and allow enterprises or partners to access the business system to establish permissions, support and feedback mechanisms.

stage	output
Pilot preparation	Scenario description, data preparation checklist, redaction instructions, and version records.
Run verification	Data mapping, target and rule configuration, baselines and tuning plans, versions and running status.
Result review	Description of bottlenecks, differences, conflicts and costs, review by responsible person and conclusions on continuation, adjustment or stopping.
Subsequent expansion	Template package, capability adaptation list, business authorization or ecosystem cooperation suggestions.

10. Ecosystem integration and cooperation methods

DecisioWorks provides common capabilities for data semantics, goals and rules, model construction and optimization, result analysis and feedback. Partners use the toolkit's standard actions, templates and interfaces to form customer-specific processes and industry applications, and leverage their customer relationship, on-site delivery, system integration and operation and maintenance capabilities. The organizational review is completed by the responsible person designated by the customer.

Role	Main responsibilities
DecisioWorks	Provides decision capability base, standardized interfaces, toolkits, model-based solving, result analysis and product support.
Software and Integration Partners	Responsible for system integration, industry applications, customer implementation, local services and long-term operation and maintenance.
consulting partner	Responsible for industry diagnosis, customer relations, management promotion, program design and results communication.
manufacturing company	Provide business data, production rules, pilot scenarios, result review and long-term use environment.
Park and platform	Organize enterprise access, scene screening, policy coordination, industrial services and ecosystem aggregation.

Cooperation process: Initial contact → Scenario confirmation → Data preparation → Operational validation → Result review → Decide whether to expand.

This division of labor keeps DecisioWorks light, stable, and reusable, while retaining sufficient value space for MES/ERP manufacturers, consulting agencies, system integrators, and manufacturing companies.

11. Release of information, data security and contact information

Version and release portal

DecisioCore source code is openly available as a reference implementation and easy to understand, expand and integrate; GOCK provides core computing capabilities through licensing. Enterprises and partners can choose the corresponding version and licensing plan based on their application needs.

Product Home: <https://qiaoyx-or.github.io/production-decision-sandbox>

Commercial Edition address: <https://gitee.com/qiaoyx-or/decisioworks>

Research Edition address: <https://gitee.com/qiaoyx-or/decisioworks-research>

The research version is used for learning evaluation, sample running, scenario preparation and non-production verification; the commercial version supports business applications, partner integration and continuous operation through corresponding authorization. Versions can be selected based on usage goals, capability requirements, and operational resources.

Local deployment and data management

The pilot only prepares data according to agreed scenarios and fields, without opening the complete ERP, MES, WMS database or internal implementation. Customers provide organized, redacted and versioned data packages for mapping orders, processes, resources, materials, goals and rules.

DecisioWorks commercial version supports local deployment and controls corresponding capabilities through authorization files; the standard local operation process does not require uploading business data to the external cloud. When enabling external AI, remote support, or third-party interfaces, access permissions, data flow, and redaction methods can be configured according to enterprise data management requirements.

Customers independently control business data, on-site rules and sensitive information; with customer consent, abstract field structures, templates and general configurations can be retained as reusable experiences.

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The specific version functions are subject to the official release notes, and the scope of authorization, service content, and rights and responsibilities of both parties are agreed upon in the formal agreement.